

Computational Physics 2 Classes No.1 - Report

Finite Difference Solution of the 2D Poisson Equation: grid refinement

Natalia Kowalczyk

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1 Introduction

The Poisson equation is a fundamental partial differential equation that appears in many problems of physics, including electrostatics, gravitation, and diffusion. In this project, the two-dimensional Poisson equation with homogeneous Dirichlet boundary conditions was solved numerically using the finite difference method on uniform grids.

The main goal was to study the convergence of three iterative approaches: standard Gauss–Seidel relaxation, successive overrelaxation (SOR), and grid refinement. The quality of the numerical solution was assessed using the residual, the energy functional $S[u]$, and comparison with the exact analytical solution.

2 Results

2.1 Relaxation (Gauss–Seidel)

Convergence of iteration The convergence behavior of the Gauss–Seidel method is shown in Fig. 1. The energy functional $S[u]$ decreases monotonically towards the analytical value

$$S_{\text{exact}} = -\left(\frac{3\pi}{2}\right)^2.$$

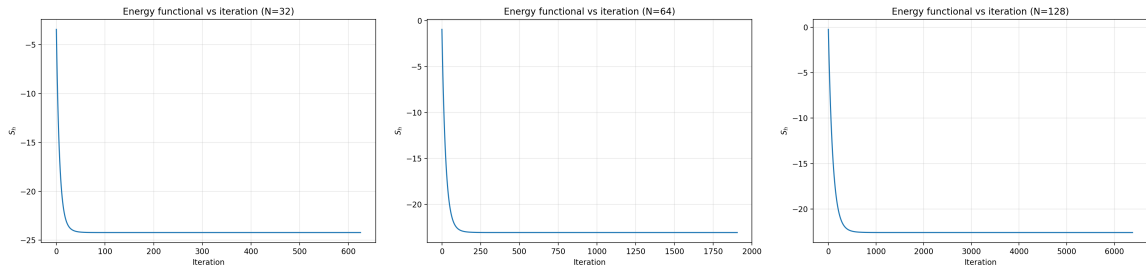


Figure 1: Energy functional vs iteration for different grid sizes.

The convergence slows significantly as N increases, which is characteristic for point relaxation methods.

Residual evolution The decay of the residual is presented in Fig. 2. Both maximum and RMS residuals decrease approximately exponentially at early stages, followed by slower asymptotic convergence.

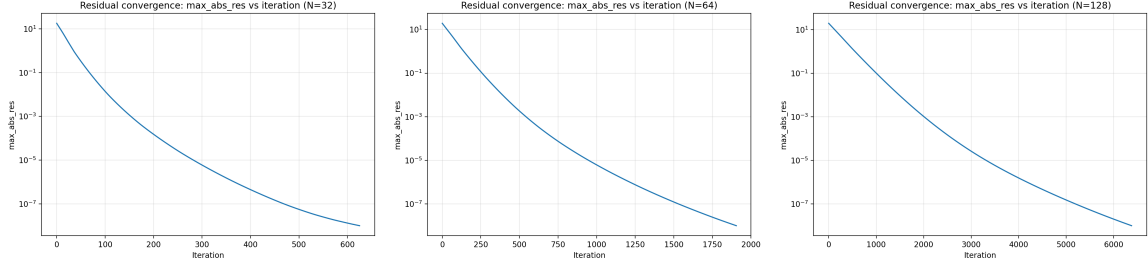


Figure 2: Maximum residual vs iteration.

Spatial structure of the residual The residual heatmaps (Fig. 3) show that the error is initially distributed across the domain and gradually concentrates in smooth long-wavelength modes.

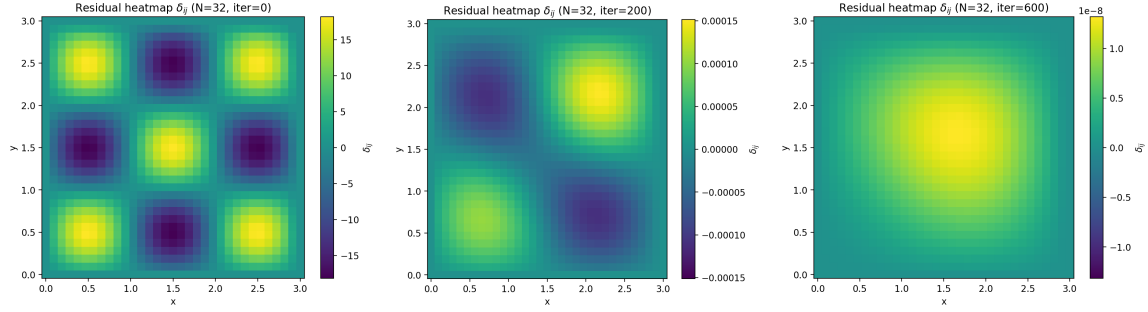


Figure 3: Residual evolution during iteration (example for $N = 32$).

Final solution The numerical solution matches the analytical solution very well, as shown in Fig. 4.

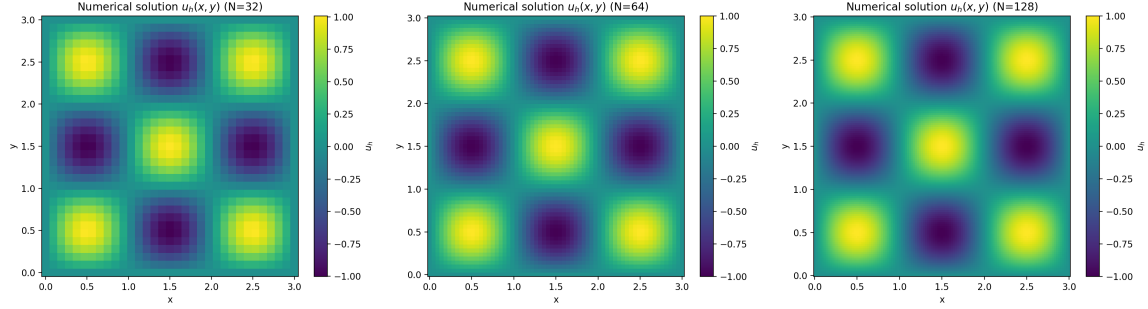


Figure 4: Numerical solution for different grid resolutions.

Convergence with grid size The error decreases with increasing N , approximately following second-order convergence, as expected for the finite difference discretization.

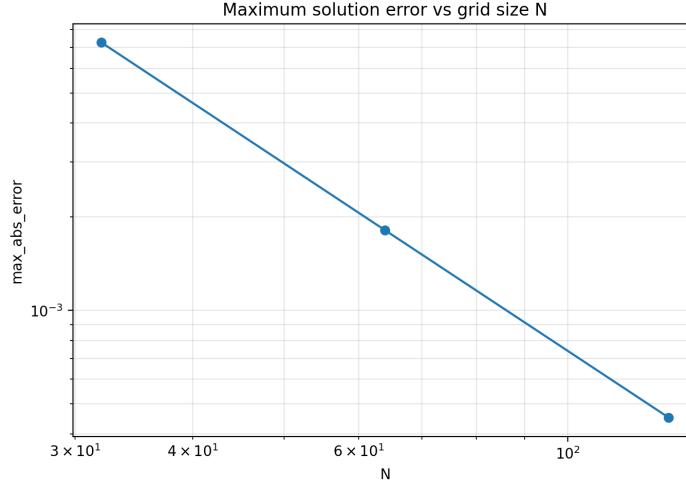


Figure 5: Maximum error vs grid size.

2.2 Overrelaxation (SOR)

Optimal relaxation parameter The optimal relaxation parameter ω was determined by minimizing the number of iterations required for convergence. The results are shown in Fig. 6.

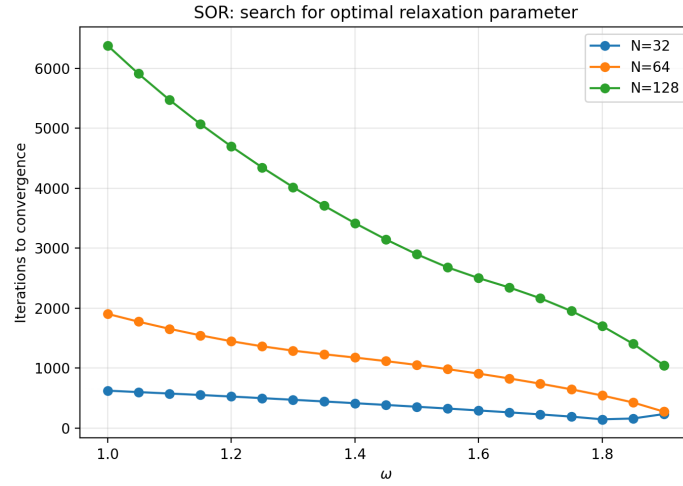


Figure 6: Number of iterations vs relaxation parameter ω .

The optimal value is found to be approximately

$$\omega_{\text{opt}} \approx 1.8 - 1.9,$$

which is consistent with theoretical expectations.

Accelerated convergence Using ω_{opt} , the convergence is significantly faster compared to standard Gauss-Seidel, as shown in Fig. 7.

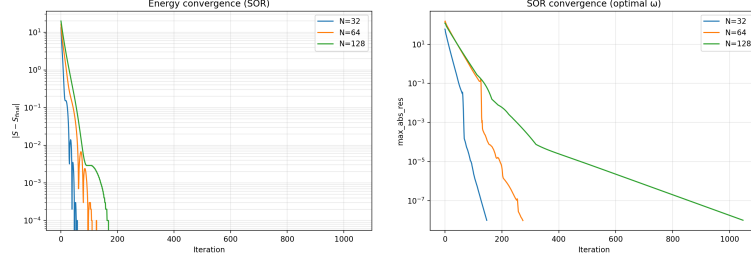


Figure 7: Convergence of SOR method.

The number of iterations is reduced by an order of magnitude.

2.3 Grid refinement

In the grid refinement approach, the solution from a coarser grid is used as an initial condition on a finer grid. Values at existing nodes are copied, while new nodes are initialized as averages of neighboring coarse-grid values.

Multilevel convergence The multilevel approach significantly improves convergence by transferring solutions between grids. The energy as a function of total iteration count is shown in Fig. 8.

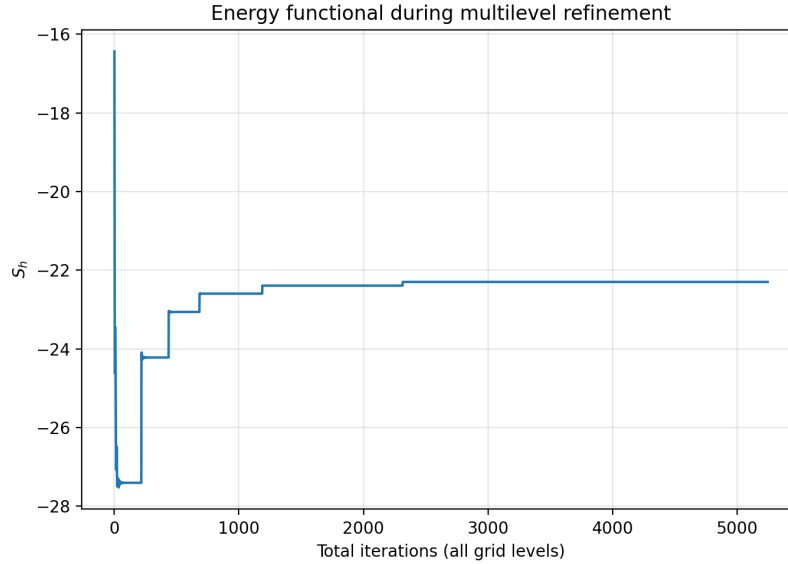


Figure 8: Energy vs total number of iterations for multilevel solver.

Maximum achievable resolution Using grid refinement, significantly larger grids can be reached within the time constraint. In this implementation, grids up to

$$N = 2^9 = 512$$

are feasible within the allowed runtime.

Efficiency improvement Compared to solving directly on the finest grid, the multilevel method reduces the total number of iterations required for convergence, demonstrating improved efficiency.

3 Conclusions

The finite difference solution of the 2D Poisson equation was successfully obtained using iterative methods. The Gauss–Seidel method was found to converge reliably but slowly, with the convergence rate deteriorating as the grid resolution increased. The residual and energy functional confirmed correct convergence towards the analytical solution.

The introduction of overrelaxation (SOR) significantly improved convergence speed. An optimal relaxation parameter $\omega \approx 1.8$ – 1.9 was identified, reducing the number of iterations by approximately an order of magnitude compared to standard relaxation.

Finally, the grid refinement (multilevel) approach further enhanced efficiency by providing good initial guesses on finer grids. This allowed larger grid sizes to be reached within reasonable computational time. Overall, the results are consistent with theoretical expectations and demonstrate the effectiveness of these methods for solving elliptic problems.